Trainings ▾

Preparatory Steps

Industrial and Power Generation

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html)

QSK60 Marine Applications

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html)
- Remove the aftercooler heat shields. Refer to Procedure 010-129 in Section 10. (/qs3/pubsys2/xml/en/procedures/56/56-010-129-tr.html)

Remove

QSK45 Engines

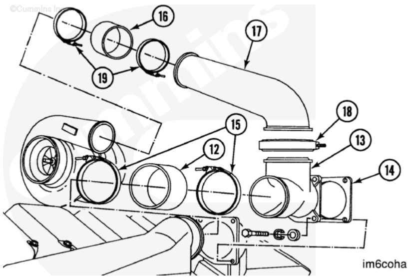
Loosen the hose clamps (19) between the air crossover (17) and the turbocharger.

Loosen the hose clamps (15) between the T-connection (13) and the aftercooler.

Remove the V-band clamp (18) between the air crossover and the T-connection. Pull the air crossover and the hose (16) from the turbocharger. Push the hose (12) on the T-connection (13) as far as possible.

Remove the four capscrews. Remove the T-connection and gasket (14). Remove the hose from the connection.

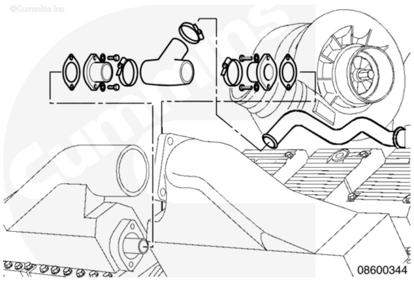
Discard the gasket.



Note : Procedures for one side of the engine are shown. Procedures for the other side are similar.

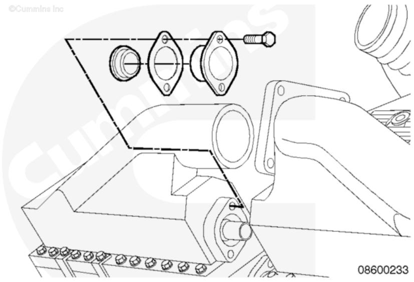
Remove the three hose clamps from the rubber T-connection.

Remove the rubber T-connection.



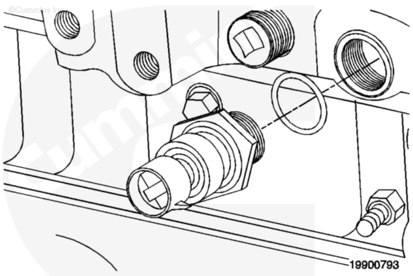
Remove the capscrews, seal retainers, gaskets, and seals from each aftercooler.

Discard the seals and gaskets.



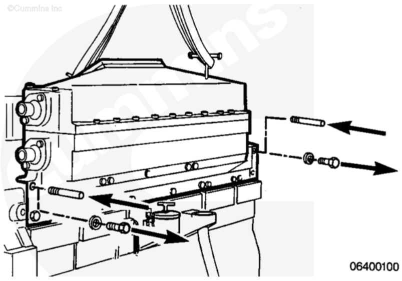
Remove the intake manifold pressure sensor.

Remove the intake manifold temperature sensor.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Remove two of the aftercooler mounting capscrews.

Install two guide studs. Use a hoist, two T-handles, and a lifting sling.

Attach the T-handles, sling, and hoist. Raise the hoist until there is tension on the sling.

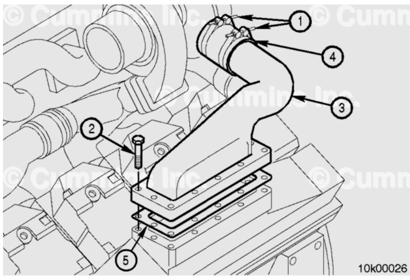
Remove the remaining ten capscrews.

Remove the aftercooler assembly.

Discard the intake gaskets.

QSK60 Engines

Note : The QSK60 engine contains an individual air crossover for each aftercooler. The configuration of the left-bank front and right-bank rear air crossovers are different from those used on the right-bank front and left-bank rear, but the removal procedures are the same.



Note : Engines using high pressure ratio compressor turbochargers use different air crossovers to verify alignment.

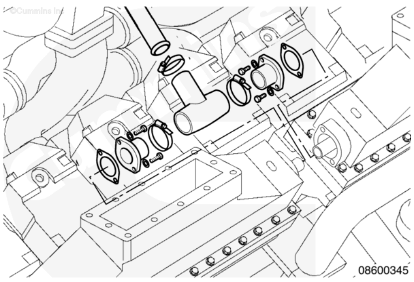
Loosen the two hose clamps (1) between the turbocharger and the air crossover (3).

Remove the twelve capscrews (2) from the air crossover.

Remove the air crossover (3), gasket (5), and hose (4) from the turbocharger and aftercooler.

Discard the gasket.

Note : Procedures for one side of the engine are shown. Procedures for the other side are similar.

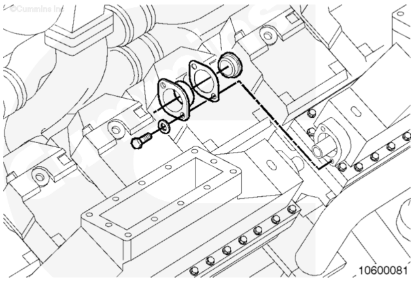


Remove the three hose clamps from the rubber tee.

Remove the rubber T-connection.

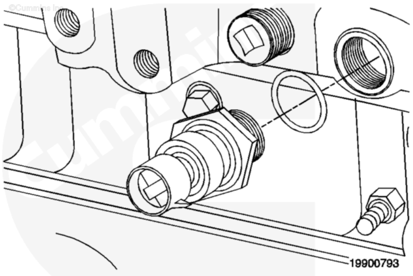
Remove the capscrews, seal retainers, gaskets, and seals from each aftercooler.

Discard the seals and gaskets.



Remove the intake manifold pressure sensor.

Remove the intake manifold temperature sensor.



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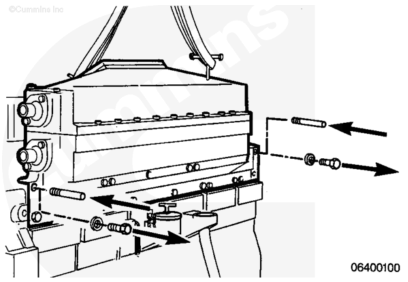


Due to the weight of the cast iron assembly, it **must** be disassembled as it is removed, if a hoist can **not** be used. Go to the Disassemble section of this procedure.

With an aluminum assembly, remove two of the aftercooler mounting capscrews.

Install two guide studs. Use a hoist, two T-handles, and a lifting sling.

Attach the T-handles, sling, and hoist. Raise the hoist until there is tension on the sling.

Remove the remaining ten capscrews

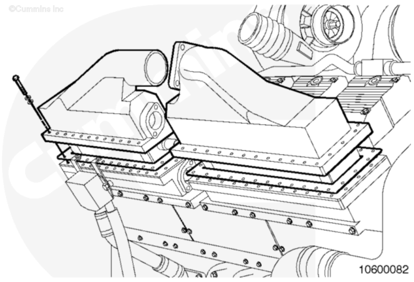
Remove the aftercooler assembly.

Discard the intake gaskets.

Disassemble

Remove the aftercooler cover capscrews, cover, and gasket.

Discard the gasket.



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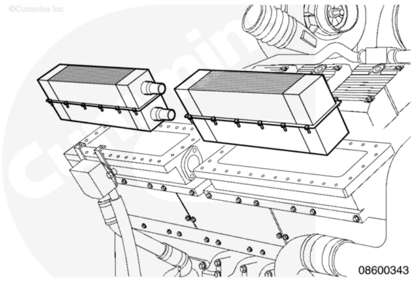


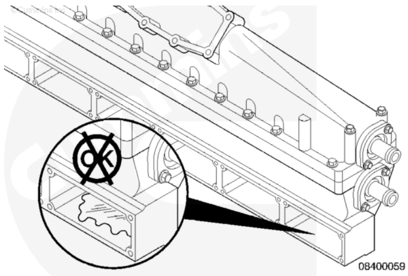
Remove the aftercooler element and gaskets.

Discard the gaskets.

Inspect for Reuse

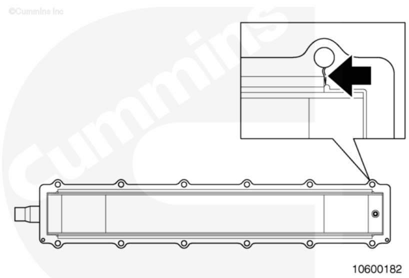
Note : After an instance of turbocharger malfunction, inspect thoroughly for debris throughout the system.

Inspect the cylinder head intake ports and the aftercooler for indications of coolant leakage.



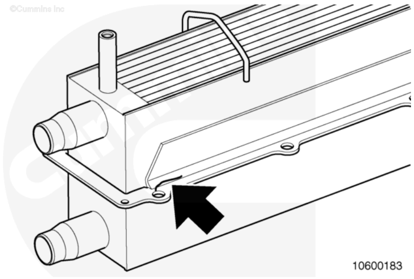
Check the aftercooler for cracks in the mounting flange at the capscrew hole area.

If the aftercooler flange is cracked, the aftercooler core **must** be replaced.



Check the aftercooler for cracks in the mounting flange.

If the aftercooler flange is cracked, the aftercooler core **must** be replaced.



Pressure Test

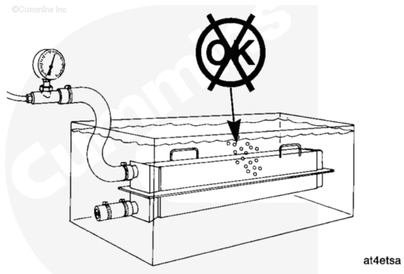
Plug one water tube with a hose that has a pipe plug installed.

Attach an air line and pressure regulator to the remaining tube.

Heating the tank of water to 50°C [122°F] improves the test results.

Place the aftercooler in a tank of water.

Apply air pressure and look for leaks.



Measurements		
	kpa	psi
Air Pressure	415	60

Do **not** attempt to repair a leaking element. The element **must** be replaced.

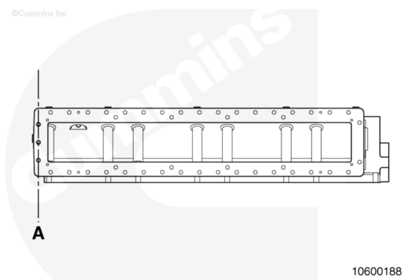
Machine

QSK60 Engines

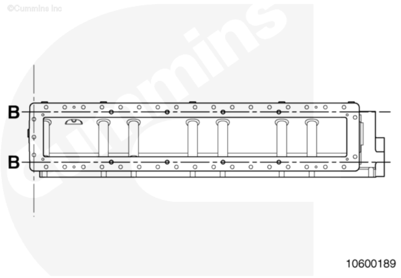
Machine the aftercooler housing and the aftercooler cover **only** if a new core is being installed.

Note : This procedure only shows one side of the aftercooler housing and aftercooler cover. The opposite side is a mirror image of the one shown. The dimensions and base line are identical.

Draw a line through the middle three capscrew holes (A) on the end of the aftercooler housing.



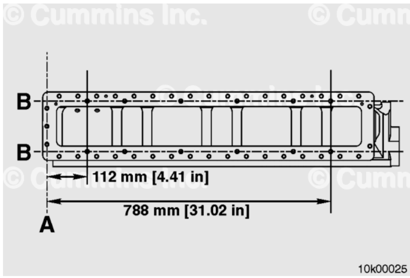
On each side of the aftercooler housing, draw a line through the middle of the aftercooler element mounting holes (B).



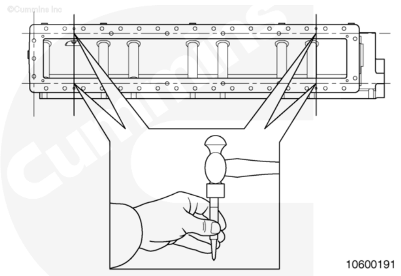
Line (A) is used as the base on both sides of the aftercooler housing.

Measure from line (B) along line (A) and mark a line at 112 mm [4.41 in] on both sides of the aftercooler housing.

Measure from line (B) along line (A) and mark a line at 788 mm [31.02 in] on both sides of the aftercooler housing.



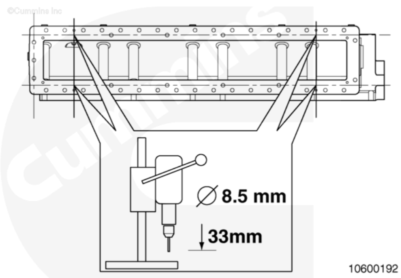
Center punch the four holes.



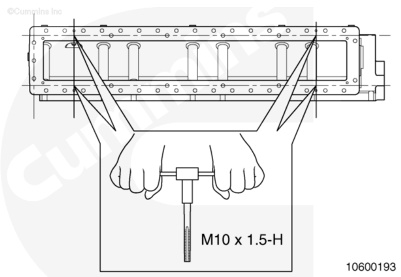
Place the aftercooler housing in a drill press.

Drill four holes using a 8.5 mm drill to a depth of 33 mm [1.3 in].

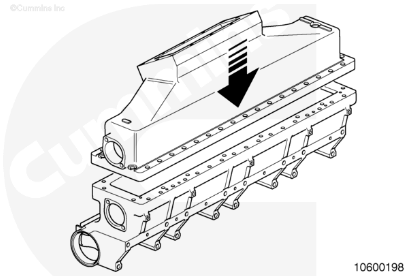
Clean out the holes.



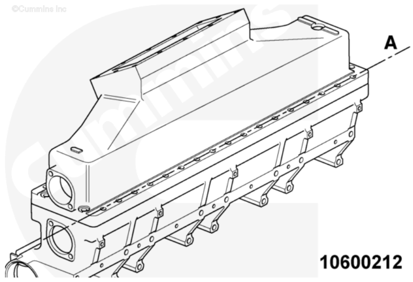
Tap the four holes with the use of a M10 x 1.5-H tap.



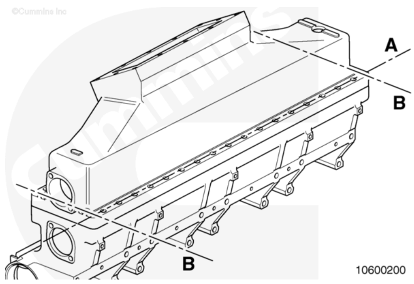
Assemble the aftercooler cover to the aftercooler housing without the aftercooler core.



Draw a line through the center of capscrew holes (A) on the side of the aftercooler cover.

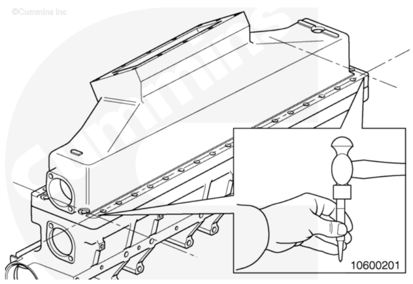


Draw a line through the center of the capscrew holes (B) on the end of the aftercooler cover.



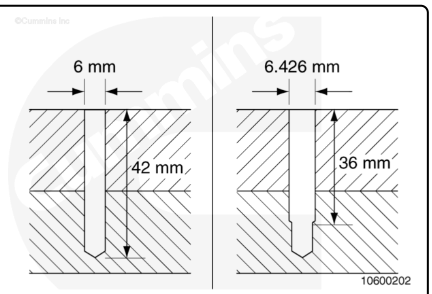
Center punch the aftercooler cover at the intersection of lines A and B.

Place the assembly in a drill press.



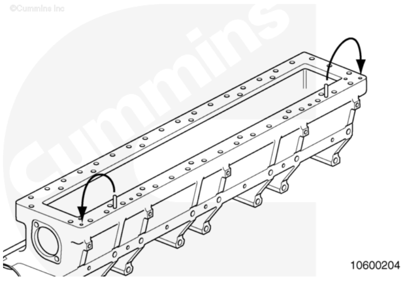
Use a 6 mm drill to drill the center punched marks to a depth of 42 mm [1.65 in].

Enlarge the holes, with a 6.4 mm [0.252 in] drill to a depth of 36 mm [1.47 in]

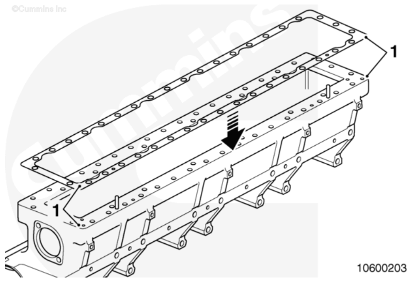


Disassemble the aftercooler cover and aftercooler housing.

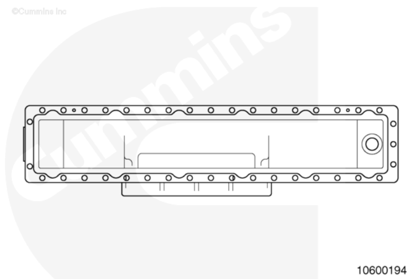
Remove the dowel pins from their present locations and install them into the newly drilled holes.



Place the new gasket (1) onto the aftercooler housing (1) and add two extra holes for the dowel pins.



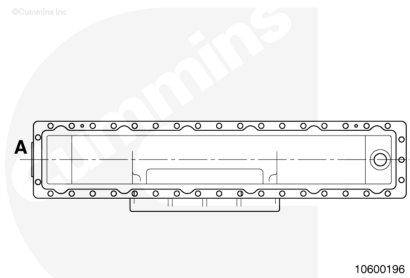
Aftercooler cover, Part Numbers 4001329 and 4001330, require four extra slots to be added to provide clearance for new aftercooler element mounting capscrews.



Base line (A) is the edge of the aftercooler cover where the aftercooler element water outlet protrudes.

Measure from base line (A) 800 mm [31.50 in] and make a mark.

Measure from base line (A) 124 mm [4.88 in] and make a mark.



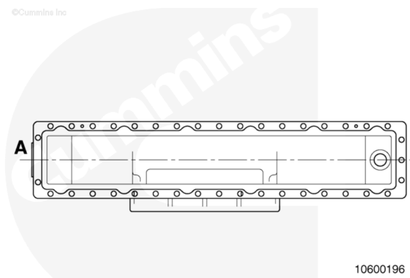
Place the aftercooler cover into a milling machine.

Use a 35 mm [1.38 in] slot drill to machine the slots in the aftercooler cover.

Align the slot faced milling drill to the center line of the aftercooler cover.

Move the slot faced milling drill to the first 124 mm [4.88 in] mark.

Move the slot faced milling drill 66 mm [2.59 in] toward the first 124 mm [4.88 in] mark.

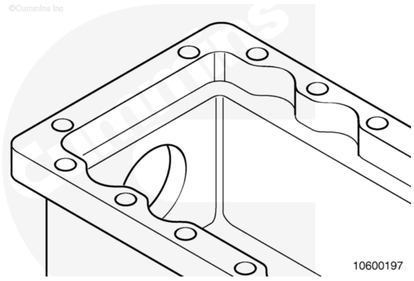


Drill a new slot to the depth of 20 mm [0.79 in].

Repeat the drilling process for the three remaining marked areas.

The new slots will be the same size as the existing twelve slots.

Remove all debris and thoroughly clean the cover.



Assemble

QSK45 Engines

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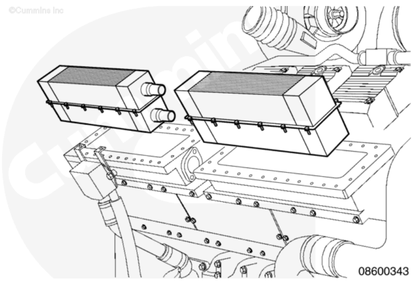
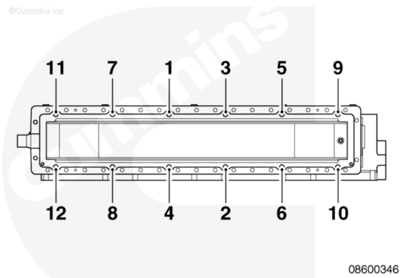


Make sure the aftercooler supports are positioned correctly.

Install the aftercooler element and top gasket.

Install the twelve capscrews and tighten in the sequence shown.

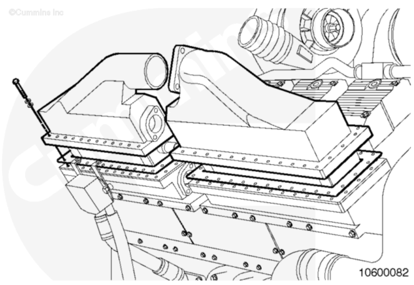
Torque Value:
Aftercooler Element Capscrews 45 n•m [33 ft-lb]



Apply a light coat of anti-seize lubricant, Part Number 3824759, to the aftercooler cover capscrew threads.

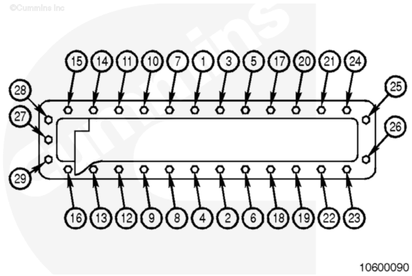
Install the aftercooler cover.

Install the twenty nine aftercooler cover capscrews.



Tighten the twenty nine capscrews in the sequence shown.

Torque Value:
QSK45 Aftercooler Cover Capscrews 45 n•m [33 ft-lb]



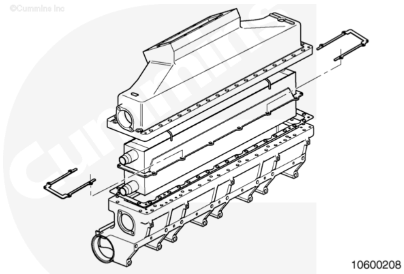
QSK60 Engines

Note : For cast iron systems, if a crane can not be used, the system must be assembled on the engine by mounting the intake manifolds first, and then assembling the aftercoolers.

If the existing aftercooler does **not** have flange cracks and can be reused, it is recommended that U-shaped clamping plates be installed.

The plates will help to lower the assembly stress and increase the strength of the aftercooler element flange.

- Use the following procedure for reuse guidelines. Refer to Procedure 010-008 in Section 10. (/qs3/pubsys2/xml/en/procedures/56/56-010-008.html)



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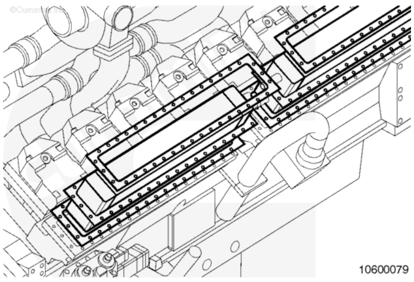
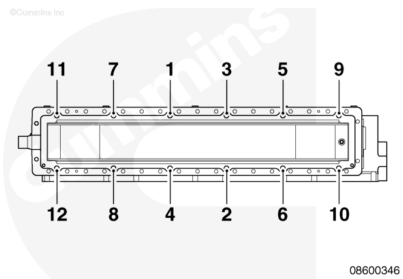
Make sure the aftercooler element supports are positioned correctly.

Install the aftercooler element and top gasket.



Install the twelve capscrews and tighten in the sequence shown.

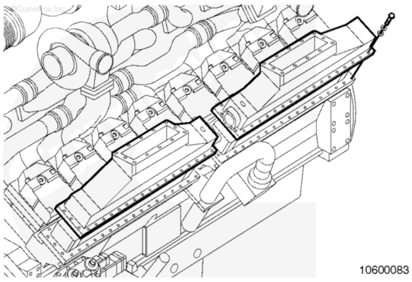
Torque Value:
Aluminum Aftercooler Assembly Element 45 n•m [33 ft-lb]



Torque Value:
Cast Iron Aftercooler Assembly Element 65 n•m [48 ft-lb]

Apply a light coat of anti-seize lubricant, Part Number 3824759, to the capscrew threads.

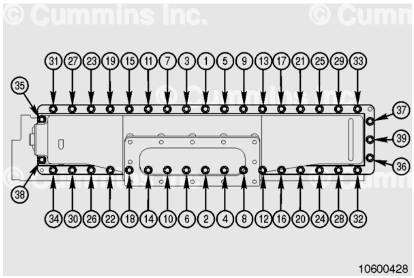
Install the aftercooler cover and capscrews.



Tighten the thirty nine aftercooler cover capscrews in the sequence shown.

Torque Value:
Aluminum Aftercooler Assembly 45 n•m [33 ft-lb]

Torque Value:
Cast Iron Aftercooler Assembly 65 n•m [48 ft-lb]



Install

QSK45 Engines

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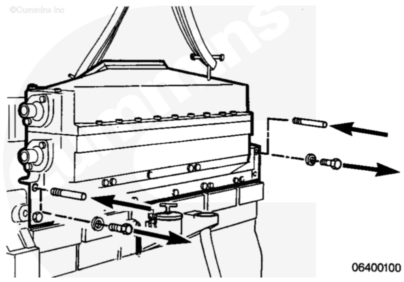


Use guide studs long enough to protrude beyond the installed part.

Install two guide studs in the lower row of capscrew holes in the cylinder heads.

Install the new intake gaskets, with the raised bead of the gasket toward the cylinder head, to the manifold with screws.

Use Lubriplate™ 105 to lubricate the aftercooler element spouts.

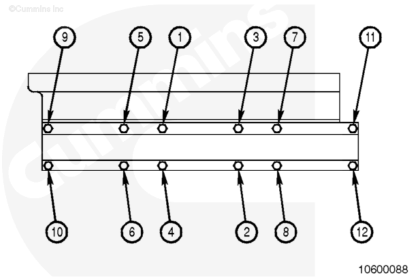
Lightly tap the aftercooler assembly into location so the core spouts are positioned in the water connector.

Install the aftercooler and the top row of capscrews.

Tighten the capscrews **only** enough to hold the assembly on the engine.

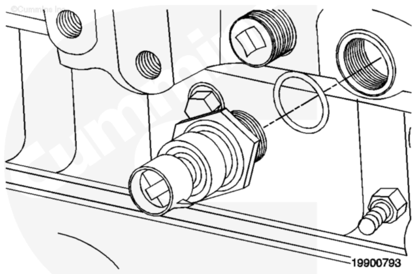
Remove the guide studs. Install the bottom row of capscrews.

Tighten the intake manifold mounting capscrews in the sequence shown.



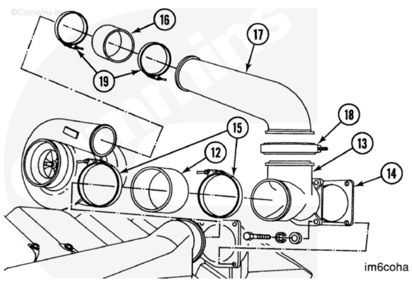
Torque Value:
QSK45 Intake Manifold Capscrews 45 n•m [33 ft-lb]

Install the intake manifold pressure and temperature sensors.



Torque Value:
Intake Manifold Pressure and Temperature Sensors 15 n•m [133 in-lb]

Position the air crossover and the hose (16) on the turbocharger. Install the V-band clamp (18) on the air crossover and the T-connection.



Install the hose clamps (15) between the T-connection (13) and the aftercooler.

Install the T-connection and gasket (14) with four capscrews. Install the hose on the connection.

Torque Value:
Aftercooler Connection T-Mounting Capscrews 45 n•m [33 ft-lb]

Install the hose clamps (19 and 15) between the air crossover (17) and the turbocharger.

Torque Value:
Intake Air Systems Hose Clamps 9 n•m [80 in-lb]

Lubricate the seal with clean engine oil.

Install the seal on the retainer.

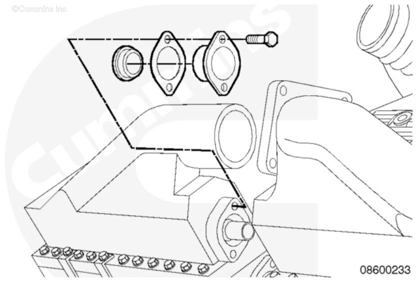
Lubricate the outer diameter of the water bosses on the aftercooler element with Lubriplate™ 105.

Slide the gasket on the water bosses.

Slide the seal and the retainer on the tube.

Install and tighten the capscrews.

Torque Value: 45 n•m [33 ft-lb]



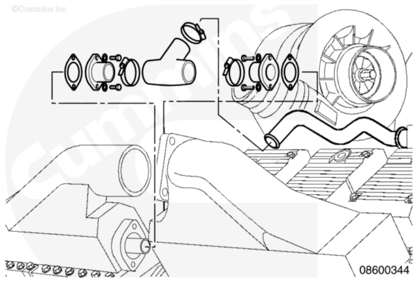
Note : Procedures for one side of the engine are shown. Procedures for the other side are similar.

Install the two rubber T-connections on the aftercooler tubes.

Install the three hose clamps on the rubber T-connections.

Tighten the hose clamps.

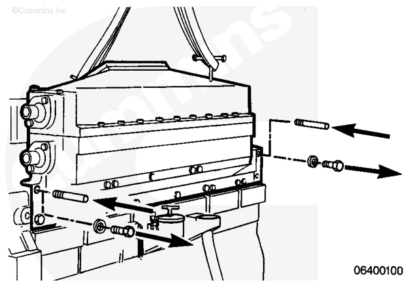
Torque Value: 5 n•m [44 in-lb]



QSK60 Engines

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Use guide studs long enough to protrude beyond the installed part.

Install two guide studs in the lower row of capscrew holes in the cylinder heads.

Install the new intake gaskets, with the raised bead of the gasket toward the cylinder head, to the aftercooler housing with screws.

Use Lubriplate™ 105 to lubricate the aftercooler element core spouts.

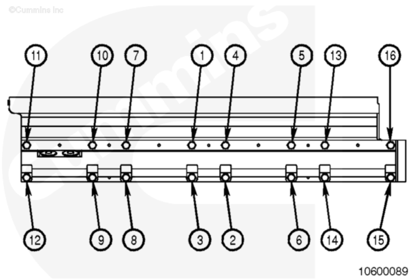
Lightly tap the aftercooler assembly into location so the core spouts are positioned in the water connector.

Install the aftercooler and the top row of capscrews.

Tighten the capscrews **only** enough to hold the assembly on the engine.

Remove the guide studs. Install the bottom row of capscrews.

Tighten the aftercooler assembly mounting capscrews in the sequence shown.



Torque Value:
Aluminum Aftercooler Assembly 45 n•m [33 ft-lb]

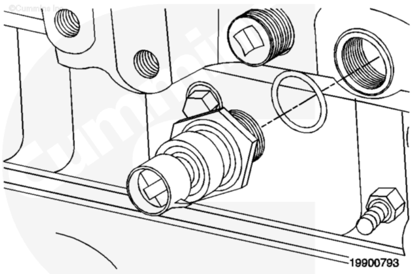
Torque Value:
Cast Iron Aftercooler Assembly 65 n•m [48 ft-lb]

After 20 to 30 minutes, tighten the aftercooler assembly capscrews again in the sequence shown.

Torque Value:
Aluminum Aftercooler Assembly 45 n•m [33 ft-lb]

Torque Value:
Cast Iron Aftercooler Assembly 65 n•m [48 ft-lb]

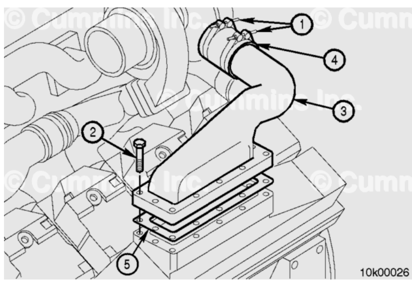
Install the intake manifold pressure and temperature sensors.



Torque Value:
Intake Manifold Pressure and Temperature Sensors 15 n•m [133 in-lb]

Note : The QSK60 engine contains an individual air crossover for each aftercooler. The configuration of the left-bank front and right-bank rear air crossovers are different from those used on the right-bank front and left-bank rear, but the installation procedures are the same.

Note : Engines using high pressure ratio compressor turbochargers use different air crossovers to verify alignment. Verify the same parts are installed as removed.



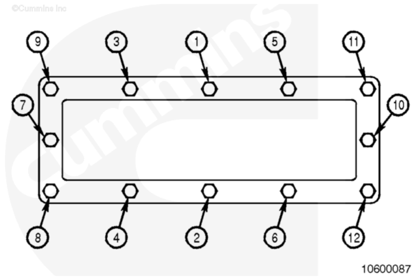
Position the air crossover, gasket, and hose on the turbocharger and aftercooler.

Install the two hose clamps (1) between the turbocharger and the air crossover (3). Verify the hump hose is installed square and even.

Torque Value:
Intake Air System Hose Clamps 9 n•m [80 in-lb]

Install and tighten the twelve capscrews on the air crossover in the pattern shown.

Torque Value:
Air Crossover Mounting Capscrews 45 n•m [33 ft-lb]



Lubricate the seal with clean engine oil.

Install the seal on the retainer.

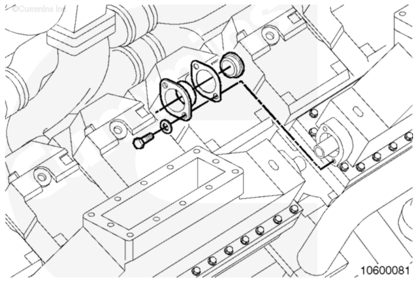
Lubricate the outer diameter of the water bosses on the aftercooler element with Lubriplate™ 105.

Slide the gasket on the water bosses.

Slide the seal and the retainer on the tube.

Install and tighten the capscrews.

Torque Value: 45 n•m [33 ft-lb]

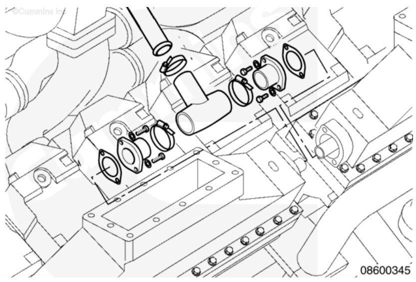


Install the two rubber T-connections on the four aftercooler tubes.

Install three hose clamps on the rubber T-connections.

Tighten the hose clamps.

Torque Value: 5 n•m [44 in-lb]



Finishing Steps

Industrial and Power Generation

- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html)
- Start and operate the engine. Check for leaks.

QSK60 Marine Applications

- Install the aftercooler heat shields. Refer to Procedure 010-129 in Section 10. (/qs3/pubsys2/xml/en/procedures/56/56-010-129-tr.html)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html)

- Start and operate the engine. Check for leaks.

Last Modified: 01-Mar-2023